Q Majority Element

→ brute force
→ T.C → O(n²)
→ S.C → O(1)

```
(1) for (i → 0 → n)
(2)   cnt → 0
(3)   for (j → 0 → n){
(4)     if (i == j) cnt++;
(5)     if (cnt > n/2) return i;
(6)   }
(7) } return -1;
```

→ Optimize (Boyer Moore Voting Algo)
→ T.C → O(n), O(1)

→ It works on the idea of vote cancellation - different element cancel each other, and the majority element remains as the final candidate.

```
cnt → 1
ele → arr[0];
for (i → 0 → n){
  if (ele == i] cnt++;
  else {
    cnt--;
    if (cnt == 0){
      ele → nums[i];
      cnt → 1;
    }
  }
} return ele;
```

Dry Run :-

[3, 2, 3, 2, 3, 3]

cnt → 1, ele → 3 → return ele

1, 0, 1, 0, 1, 2, 3